



Prevalence, Risk Factors, and Antimicrobial Resistance of *Staphylococcus* and *Streptococcus* Species Isolated from Subclinical Bovine Mastitis

Abu Al Farabi,^{1,†} Hemayet Hossain,^{2,†} Khadiza A. Brishty,^{3,†} Md Hafizar Rahman,⁴ Mostafizor Rahman,⁵ Md Saiful Islam Siddiqui,² Mohammad Atikuzzaman,⁶ Asmaa Saleh,⁷ Reem Binsuwaidan,⁷ Heba Mohammed Refat M. Selim,⁸ Ayman Noreddin,⁹ Yosra A. Helmy,^{10,11} Md Mahfujur Rahman,¹² Himel Barua,¹ and Mohamed E. El Zowalaty⁹

Abstract

Subclinical mastitis (SCM) is a prevalent serious disease among dairy cows worldwide. It poses a significant challenge to the dairy industry, animal welfare, and a threat to public health. The present study aimed to investigate the molecular detection, prevalence, and antimicrobial resistance of *Staphylococcus* spp. and *Streptococcus* spp. isolated from raw composite milk samples obtained from SCM dairy cattle in Bangladesh. A total of 612 quarters milk samples obtained from 153 cows were analyzed for SCM using the California Mastitis Test. Bacterial isolation and identification were carried out and bacterial species were confirmed using molecular polymerase chain reaction methods. Antimicrobial susceptibility testing was performed using disc diffusion method. The findings revealed that the prevalence of SCM was 70.3% (26/37), 35.95% (55/153), and 23.04% (141/612) in the herd, cow, and quarter levels, respectively. Among the positive samples, 92.7% (51/55) were *Staphylococcus* spp. (*S. aureus*, *S. chromogenes*, and *S. simulans*) and the remaining isolates were 7.3% (4/55) *Streptococcus* spp. (*Streptococcus agalactiae* and *Streptococcus dysgalactiae*). The most prevalent species was *S. chromogenes*, accounting for 67.3% (37/55). Antimicrobial susceptibility testing showed that 65.5% of isolates were susceptible to cefoxitin, whereas, 89.1% were resistant to penicillin. Overall, 12 isolates (21.8%) out of 55 were resistant to more than three classes of antimicrobials and were defined as multidrug-resistant isolates. Methicillin-resistance gene was detected in 61.1% of the cefoxitin-resistant isolates. A multivariate logistic regression analysis identified five potential risk factors including the lack of post-milking teat disinfection (OR: 3.06), absence of immediate feeding after milking (OR: 9.81), poor udder hygiene (OR: 7.83), tick infestation (OR: 13.76), and absence of dry cow therapy (OR: 3.31). The findings of the current study underscore the urgent requirement for targeted interventions, considering the identified factors to effectively manage and control SCM in dairy cows.

¹Department of Microbiology and Veterinary Public Health, Chattogram Veterinary and Animal Sciences University, Chittagong, Bangladesh.

²Faculty of Veterinary, Animal and Biomedical Sciences, Sylhet Agricultural University, Sylhet, Bangladesh.

³Department of Zoology, University of Dhaka, Dhaka, Bangladesh.

⁴Department of Pathology and Parasitology, Chattogram Veterinary and Animal Sciences University, Chittagong, Bangladesh.

⁵Department of Dairy & Poultry Science, Hajee Mohammad Danesh Science and Technology University, Dinajpur, Bangladesh.

⁶Department of Surgery and Theriogenology, Faculty of Veterinary, Animal and Biomedical Sciences, Sylhet Agricultural University, Sylhet, Bangladesh.

⁷Department of Pharmaceutical Sciences, College of Pharmacy, Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

⁸Department of Pharmaceutical Sciences, College of Pharmacy, AlMaarefa University, Riyadh, Saudi Arabia.

⁹Department of Microbiology and Immunology, Faculty of Pharmacy, Ahram Canadian University, Giza, Egypt.

¹⁰Department of Veterinary Science, Martin-Gatton College of Agriculture, Food, and Environment, University of Kentucky, Lexington, Kentucky, USA.

¹¹Department of Zoonoses, Faculty of Veterinary Medicine, Suez Canal University, Ismailia 41522, Egypt.

¹²Department of Medicine, Faculty of Veterinary, Animal and Biomedical Sciences, Sylhet Agricultural University, Sylhet, Bangladesh.

[†]These authors contributed equally.

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Introduction

Bovine mastitis is a frequent and highly prevalent infectious disease in dairy cattle. It remains a major challenge for global dairy industries. This complex disease in dairy cows poses significant risks to public health and dairy industry worldwide, when unpasteurized milk is consumed (Paramasivam et al., 2023). Since milk has high microbial populations, the risk of antimicrobial residues and compounds derived from an immune response and inflammation process may lead to food safety concerns (Paramasivam et al., 2023; Schadt, 2023). The disease is characterized by physical, chemical, and microbial alterations in the milk, as well as structural changes in the glandular tissue, which affect the normal flow and grade of the milk (Tezera and Aman Ali, 2021). Mastitis can be divided into clinical and subclinical (SCM) form with global prevalence rates of 42% and 15%, respectively (Krishnamoorthy et al., 2021; Paramasivam et al., 2023). Subclinical mastitis is a form of mastitis in dairy cows where there are no visible signs of inflammation or illness in the udder or milk, but there is an increase in somatic cell count (SCC) and/or the presence of pathogenic bacteria in the milk (Abebe et al., 2016). Despite its public health implications, such as milk-based foodborne illness and gastrointestinal symptoms, mastitis has a profound economic impact on the dairy industry globally, mainly by reducing milk production, milk quality, retail value, and increasing treatment and labor costs (Guimarães et al., 2017). However, as with other infectious diseases, the risk factors associated with mastitis primarily depend on three key elements: exposure to microorganisms, cow immune system, and environmental and maintenance factors (Tezera and Aman Ali, 2021). Previous investigations evaluating risk factors for mastitis included age, lactation stage, milk production, breed, and body condition score (Busato et al., 2000; Oliveira et al., 2015; Sarker et al., 2013). Various microorganisms are significant etiological agents of mastitis (Ferdaus et al., 2019; Kakooza et al., 2023). Among the causative bacteria, different strains of *Staphylococcus* spp. and *Streptococcus* spp. are more frequent and transmissible from animals to humans (El-Sayed et al., 2017; Juhász-Kaszanyitzky et al., 2007).

The widespread use of antimicrobials in dairy cows to combat mastitis plays a key role in the emergence of antimicrobial resistance (AMR) in dairy cows (Rafif Khairullah et al., 2022). There were no distinctive AMR patterns observed among the typical microorganisms associated with causing SCM in Bangladesh (Chowdhury et al., 2024; Rana et al., 2022).

As mastitis is a global issue both for dairy industry and public health, it requires extensive investigation to develop strategies to pull the reins on its potential threats (Hoque et al., 2023). Therefore, the present study aimed to determine the current prevalence of SCM using molecular methods and AMR pattern, as well as assessing the associated risk factors

by building a multilevel regression model at particular regions in Chattogram, Bangladesh.

Materials and Methods

Ethics statement

Ethical Approval Committee of Chattogram Veterinary and Animal Science University (CVASU), Bangladesh approved both field and laboratory procedures of the current study [Approval no. CVASU/Dir (R&E) EC/2022/023].

Study area and design

A cross-sectional study was conducted in 11 upazilas (sub-district) of Chattogram district from January to April 2023, as shown in Figure 1. The inclusion criteria of the farms in the study were as follows: the farm should have a population of dairy cows exceeding five, a minimum establishment period of three years, and each cow should yield a minimum of 5 L of milk.

Determination of sample size

A combination of cluster and random sampling scheme was incorporated for the selection of study farms and animals. The desired criteria-based herd list was compiled and organized by sub-district categories. A minimum of two herds and ten milch cows (each herd is considered as a cluster) from each sub-district were selected randomly. The minimum number of sample size was estimated according to the formula of Al Naser et al., (2024) and Rahman et al., (2024). Following the calculation, a total of 384 milk samples were determined as necessary for estimating SCM. The study was then conducted using 612 milk samples collected from 153 cows across 37 herds.

Data and sample collection

A customized version of the Kobo Tool Box (accessible at www.kobotoolbox.org) was utilized for the comprehensive collection of explanatory, predictor, and risk factor-associated data pertaining to multiple dimensions: animal well-being, husbandry practices, and individual animal characteristics (Supplementary Data S1). Data were carefully collected through a combination of well-structured, pre-tested questionnaire encompassing both open and closed questions, thorough on-site observations, and meticulous assessments of the farm environment. The variables were carefully selected through analysis of previous studies with modifications (Hoque et al., 2023; Rana et al., 2022; Sarker et al., 2013).

Milk sampling was carried out following aseptic procedures as described by National Mastitis Council (NMC, 2004). For sampling, approximately 2 mL of milk was collected using the stripping milking technique from the cow which appeared in a good health. The process of sample collection involved an initial step of cleansing then dried the teat,

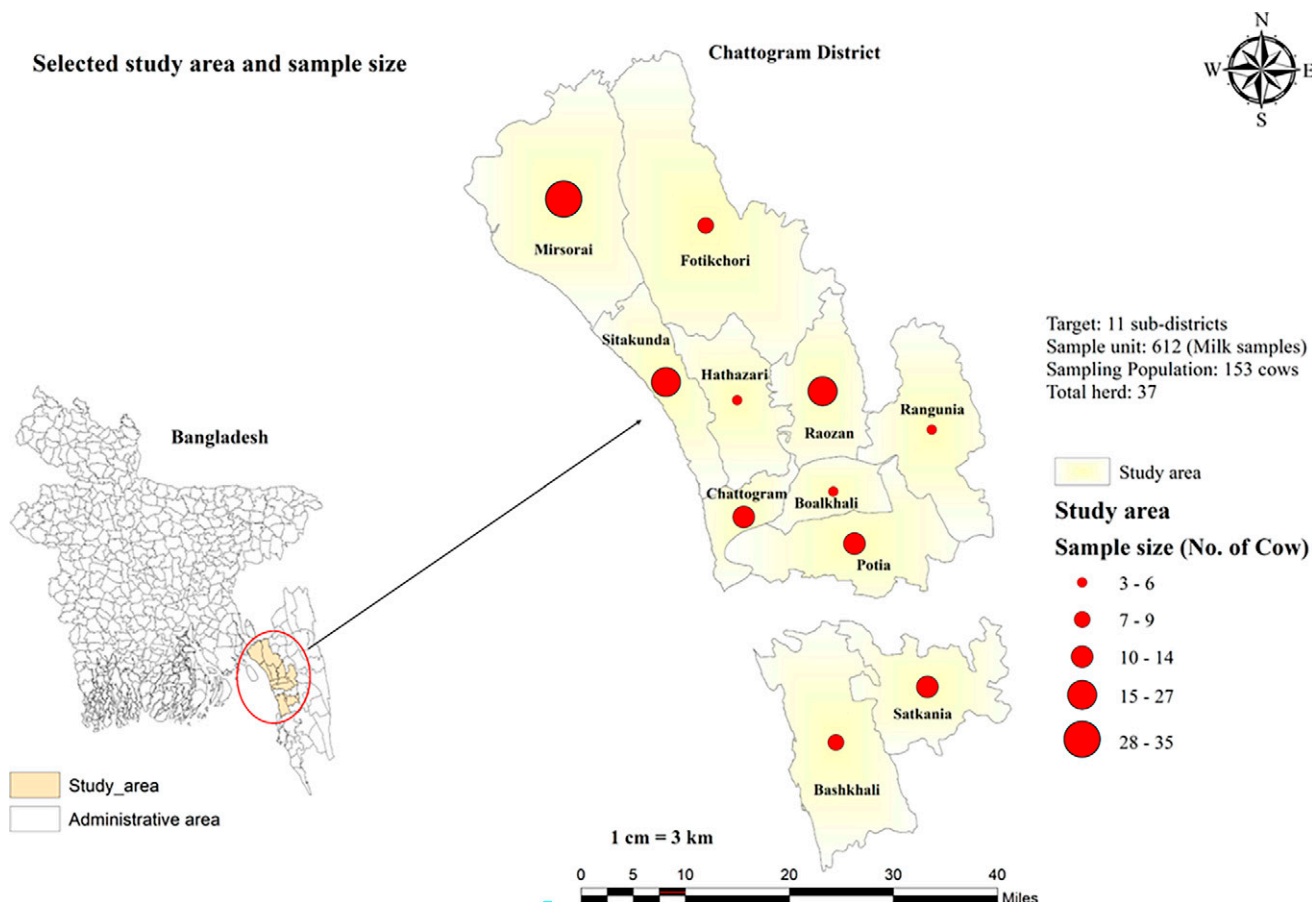


FIG. 1. Geo-spatial map showing the locations and sample size in the present study. The map was created using ArcMap 10.7. (ESRI, CA, USA).

followed by discarding the initial milk drops. The teat orifices were further cleaned by gently swabbing with a cotton ball soaked in 70% ethanol.

Disease screening by California mastitis test (CMT) and bacteriological methods

The California mastitis test (CMT) was used as a screening test for subclinical mastitis. The CMT was performed for each quarter as previously described by Abebe et al., (2016). If the CMT test showed positive results, then the samples were amalgamated to form a composite sample (Al Emon et al., 2024) for identifying selected pathogens according to the sampling process of the NMC guidelines (NMC, 2004). In the present study, the isolation and identification of *Staphylococcus* and *Streptococcus* spp. from milk samples involved both culture and biochemical techniques along with molecular detection methods using polymerase chain reaction (PCR). For *Staphylococcus* spp., 1 mL of the milk sample was enriched in 9 mL of Tryptic Soy Broth and incubated at 37°C for 24 h. A loopful of the enriched broth was streaked onto Mannitol Salt Agar (Oxoid, Cambridge, UK) plates and incubated at 37°C for 24 h. Yellow colonies indicative of mannitol fermentation was observed, suggesting the presence of *Staphylococcus* spp. Suspected colonies were subcultured onto Blood Agar plates for further isolation and incubated at 37°C for 24 h. Hemolysis patterns were examined, followed by Gram staining, which

revealed Gram-positive cocci in clusters. A catalase test was performed using 3% hydrogen peroxide, and the presence of bubbling confirmed a positive result, identifying the isolates as *Staphylococcus* spp.

For the isolation of *Streptococcus* spp., after a similar pre-enrichment, a loopful of the broth was streaked onto Blood Agar plates and incubated at 37°C for 24–48 h. The plates were examined for beta-hemolytic colonies, indicative of pathogenic *Streptococcus* spp. Gram staining of suspected colonies revealed Gram-positive cocci arranged in chains. A catalase test was conducted, and the absence of bubbling confirmed a negative result, consistent with *Streptococcus* spp. Finally, molecular identification using PCR was conducted to confirm the species-level identification of the isolates as previously reported by Hoque et al., (2023) and Rana et al., (2022).

DNA extraction and PCR

Freshly cultured bacterial isolates were used for chromosomal DNA isolation using the boiling method as previously described by Hoque et al., (2023). Species-specific primers targeting different *Staphylococcus* and *Streptococcus* spp. were used for the molecular detection as previously described by Shome et al., (2011) as shown in Supplementary Table S1. All genomic DNA and the PCR confirmed bacterial isolates were stored at –80°C for further analysis.

Antimicrobial susceptibility testing

To determine the AMR profiling, disk diffusion method was used according to the guidelines of Clinical and Laboratory Standards Institute (CLSI, 2020). Eight different antimicrobial agents from six different antibiotic classes were used, namely ampicillin (AMP) 10 µg, cefoxitin (FOX) 30 µg, ciprofloxacin (C) 5 µg, erythromycin (E) 15 µg, gentamicin (CN) 10 µg, penicillin G (P) 10 µg, trimethoprim-sulfamethoxazole 1.25/23.75 µg, and tetracycline (TE) 30 µg. The zone of inhibition was measured in (mm) and results were interpreted according to the CLSI (2020) guidelines. Isolates showing resistance against at least three different classes of antimicrobial agents (≥ 3) were defined as multi-drug resistant (MDR) as previously reported by Magiorakos et al., (2012).

Detection of methicillin-resistance (*mecA*) gene

Detection of *mecA* gene was performed using PCR as previously reported by Fazal et al., (2023). Nuclease-free water was used as a negative control and a previously confirmed methicillin-resistant *S. aureus* isolate was used as a positive control.

Multiple antibiotic resistance (MAR) index

The MAR index was calculated and interpreted according to Mahen et al., (2024) using the formula:

$$\text{MAR index} = \frac{\text{The number of antibiotics to which an isolate was resistant}}{\text{The total number of antibiotics tested}}$$

MAR index value ranged from 0 to 1. Close to zero indicated highly susceptible and near 1 indicated extremely resistance.

Statistical analysis

Both the parametric and nonparametric data were coded, sorted, and thoroughly checked to minimize errors. Following this, the data were analyzed for multivariate regression and Spearman's correlation using SPSS v26 (IBM Corporation, USA) and R-Studio (Version 4.2.1), respectively. Prevalence was estimated using following formula of Asha et al., (2024).

The precessions of this estimate were ensured by 95% confidence interval (CI) estimated using Binomial exact calculation. All the relevant predictor variables associated with the response variable were chosen for univariate analysis. The univariate and multivariate logistic regression process was fully followed by Hoque et al. (2018). To evaluate the "goodness-of-fit" of the final model, the susceptibility was assessed using the Hosmer–Lemeshow test. The regression coefficients were converted into odds ratios (OR; $e\beta$). Finally,

the results were reported following multivariate regression reporting guidelines (Leibovici et al., 2020).

Plot and geo-spatial mapping

The Pearson's correlation coefficient (r^2) was calculated, and its significance was assessed using the "corrcoef" function. The resulting coefficient values were then visualized using the "metan" package in R and RStudio (Version 4.3.2). The sample size dot map and choropleth map for prevalence were constructed using ArcMap 10.7 (ESRI, CA, USA).

Results

The demographic characteristics of purposively selected cows ($n = 153$) were depicted as different category and frequency of mastitis positive, as detailed in Supplementary Table S2.

Prevalence of SCM

In the present study, SCM was detected based on the results of the CMT. The overall prevalence of SCM at herd level, animal level, and quarter level were 70.3% (26/37), 35.95% (55/153), and 23.04% (141/612), respectively (Table 1). In cases, where a cow exhibited positive results for SCM in each quarter, it was still considered as a single positive instance at the animal level. Out of 153 cows, 64.1% (98/153) showed CMT score of zero as negative (presumptive SCC < 0.2 million), whereas 11.1% (17/153) cows showed CMT score of three (presumptive SCC > 5 million), 8.5% (13/153) showed CMT score of two (presumptive SCC ranged 1.2–5 million), 13.1% (20/153) cows showed CMT score of one (presumptive SCC ranged 0.4–1.2 million). Moreover, 3.3% (5/153) were considered as trace (presumptive SCC 0.2–0.4 million).

The Holstein Friesian breed exhibited a higher prevalence (41.5%) compared to other breeds. Regarding age categories, older animals (52.6%) displayed a greater susceptibility to SCM in comparison to other adult-young cows. However, the occurrence of mastitis did not show any notable variation based on the parity, lactation stage, and others explanatory and predictor variables (Supplementary Table S3). The cow's factor associated with herd showed the cows, which were reared in muddy flooring were more likely to have SCM than brick and concrete flooring. On organism level, *S. chromogenes* showed significantly ($p < 0.001$) the highest positive rates among the SCM positive isolates (67.3%; 37/55) as shown in Supplementary Table S4. The location-based prevalence was visualized on geo-spatial mapping (Fig. 2).

TABLE 1. PREVALENCE OF SUBCLINICAL MASTITIS IN MILCH COW ESTIMATED FROM MILK SAMPLES ($N = 612$) OBTAINED FROM 153 RANDOMLY SELECTED COWS IN 14 UPAZILA IN CHATTOGRAM DISTRICT IN THE PRESENT STUDY

Level of screening	Prevalence (No. of positive samples)	95% CI	p value
Herd level	70.27% (26/37)	53.02–84.13	<0.001
Animal (Cow) level	35.95% (55/153)	28.36–44.09	
Quarter level	23.04% (141/612)	19.76–26.58	

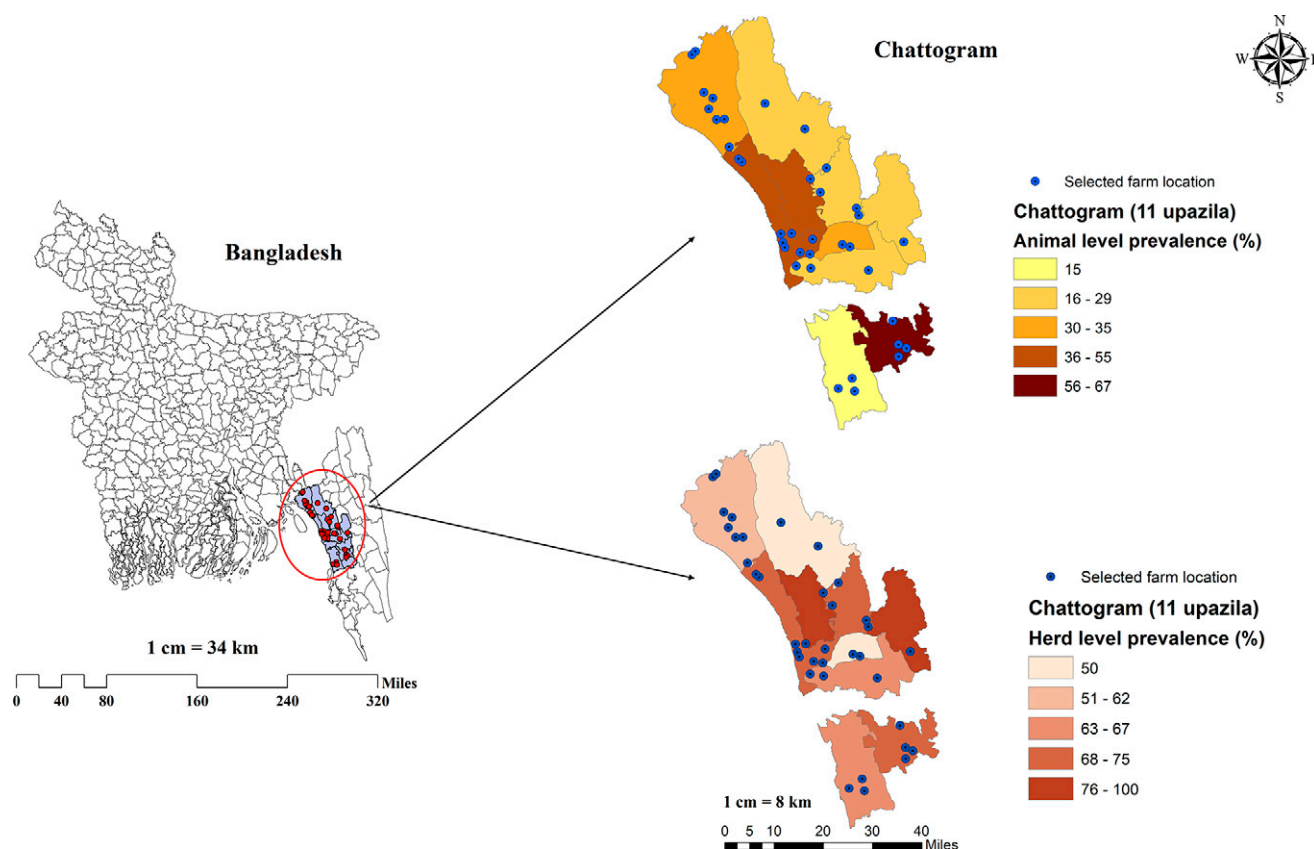


FIG. 2. Choropleth mapping showing location-based prevalence of SCM in 11 sub-districts of Chattogram, Bangladesh. The location of the selected farms was shown by XY-coordination mapping. SCM, subclinical mastitis.

Epidemiologically plausible risk factors

The outcomes of the univariate screening of explanatory, predictor variables, and epidemiologically plausible risk factors using Chi-square (χ^2) analysis were shown in Supplementary Tables S2 and S3. After screening, seven variables with p values ≤ 0.20 were included in multivariable logistic regression (MLR) analysis. However, the MLR

modeling revealed that the likelihood of SCM increased with: no post milking teat disinfection (OR = 3.06, $p = 0.006$), no feeding immediately after milking (OR = 9.81, $p < 0.001$), udder hygiene (OR = 7.83, $p < 0.001$), tick infestation on the udder (OR = 13.76, $p < 0.001$), and no dry cow therapy (OR = 3.31, $p = 0.001$), which were identified as biologically plausible risk factor of SCM (Table 2).

TABLE 2. MULTIVARIATE LOGISTIC REGRESSION ANALYSIS OF BIOLOGICALLY PLAUSIBLE RISK FACTORS OF SUBCLINICAL MASTITIS IN THE PRESENT STUDY

Independent variable	Category	OR	95% CI	p value	LRTS p value
Post milking teat disinfection	Dipping	Ref.			0.01
	Spray	0.281	0.09–0.83	0.022	
	No	3.06	1.39–6.76	0.006	
Feeding immediately after milking	Yes	Ref.			<0.001
	No	9.81	4.43–21.71	<0.001	
Body condition score	Score 5	Ref.			0.03
	Score 4	0.22	0.06–0.80	0.02	
	Score 3	0.19	0.05–0.68	0.01	
	Score 2	0.17	0.04–0.79	0.02	
	Score 1	—	—	—	
Udder hygiene	Free from dirt	Ref.			<0.001
	Slightly dirty	2.58	1.15–5.82	0.02	
	Mostly covered in dirt	7.83	2.88–21.34	<0.001	
Tick infestation on the udder	No	Ref.			<0.001
	Yes	13.76	5.76–32.84	<0.001	
Dry Cow therapy	Yes	Ref.			0.001
	No	3.31	1.66–6.59	0.001	

OR, Odds ratio, LRTS, Likelihood Ratio Test Statistics.

The likelihood ratio test statistics (LRTS) were also significant ($p < 0.05$) for these risk factor. The “Hosmer–Lemeshow goodness-of-fit test” statistic ($\chi^2 = 3.15$, $p = 0.87$) was nonsignificant, indicating the model was good fitted for the data.

The correlation matrix plot revealed the interconnectedness of various risk factors, indicating their joint influence on SCM. The majority of the risk factor were positively correlated with each other as shown in Figure 3. The body condition score, udder hygiene, “no post milking teat disinfection,” and “no/delayed feeding immediately after milking” showed moderately positive correlation with the occurrence of SCM (light blue), whereas tick infestation on udder showed negative correlation (light red).

Occurrence of etiologic agents

Out of 55 pooled SCM positive samples, it was found that *Staphylococcus* spp. (*S. aureus*, *S. chromogenes*, *S. simulans*) were identified in 92.7% (51/55) of the samples and *Streptococcus* spp. (*Streptococcus agalactiae* and *Streptococcus dysgalactiae*) were identified in 7.3% (4/55) of the samples, whereas *S. uberis*, *S. epidermidis*, and *E. coli* were not detected in the present study. Among the identified *Staphylococcus* spp., the most prevalent species was *S. chromogenes*, accounting for 67.3% (37/55; 95% CI: 53.29–79.32) of the cases (Supplementary Table S4). Conversely, the lowest frequency was observed in *S. agalactiae*, accounting for 1.8% (1/55; 95% CI: 0.05–9.72).

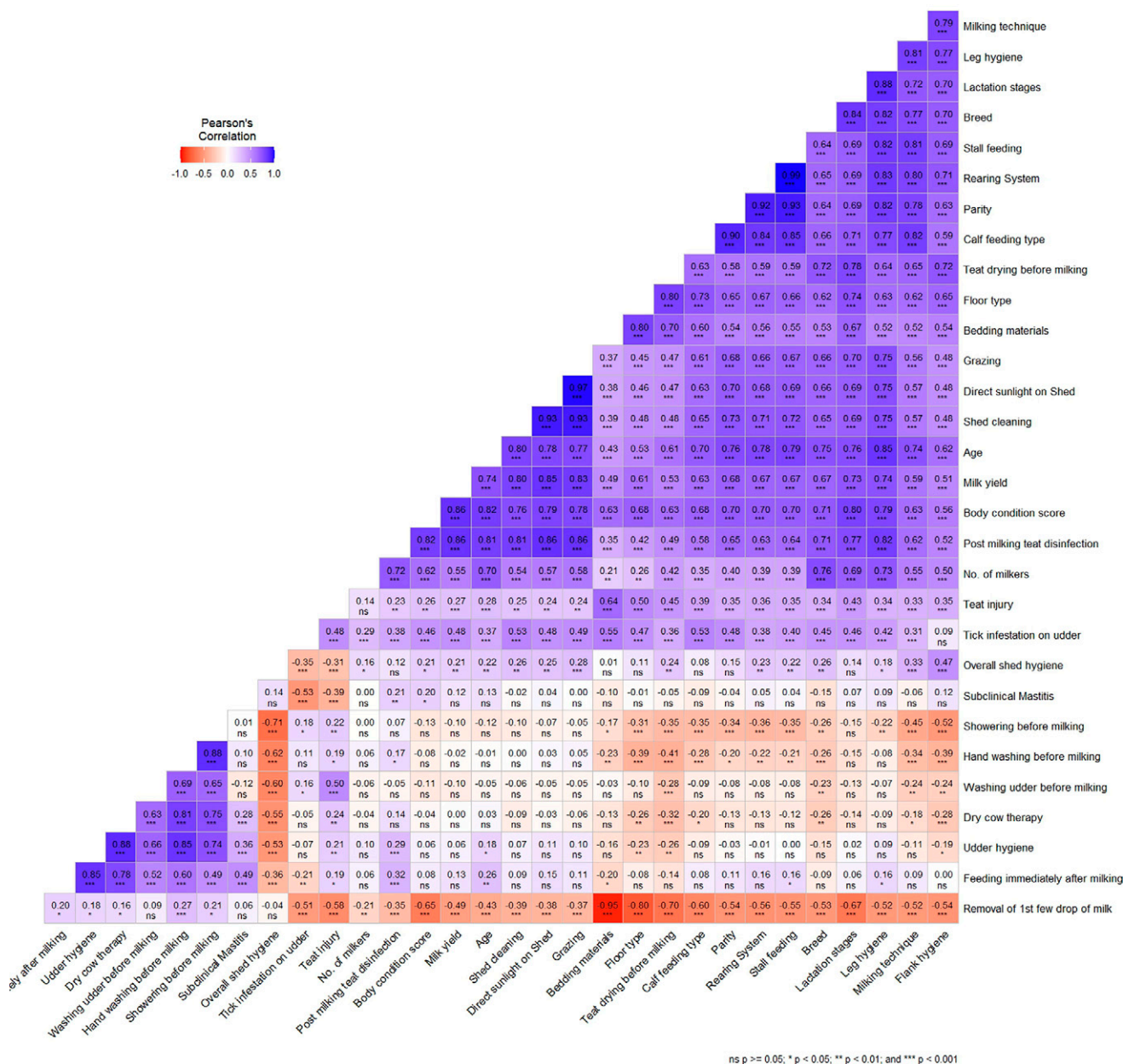


FIG. 3. Pearson correlation matrix plot showing the interconnectedness of various risk factors, indicating their joint influence on SCM in the present study. SCM, subclinical mastitis.

Antimicrobial susceptibility testing

Among the nine *S. aureus* isolates, a notable pattern emerged in their antibiotic susceptibility. They exhibited complete resistance (100%) to ampicillin and penicillin. Conversely, ciprofloxacin demonstrated the highest susceptibility (77.8%). For *S. chromogenes*, a significant percentage of the isolates exhibited susceptibility, particularly with ceftiofur (67.6%). Interestingly, *S. simulans* displayed high levels of susceptibility to the majority of antibiotics. *S. simulans* and *Streptococcus dysgalactiae* isolates showed 100% susceptibility to ceftiofur, ciprofloxacin, and tetracycline, whereas they showed 100% resistance to ampicillin and penicillin. The detailed resistance profiles of *Staphylococcus* spp. and *Streptococcus* spp. were shown in Table 3.

MDR and MAR profiles

The MAR index values ranged from 0.25 (resistant to at least two antibiotics) to 0.75. One isolate was resistant to six out of eight selected antimicrobial agents, and three isolates were resistant to five antimicrobial agents. Overall, 12 (21.8%) out of 55 isolates were resistant to more than three different categories of antimicrobial agents and were defined as MDR. Extensively drug-resistant or pan-drug-resistant isolates were not observed in the present study.

Distribution of *mecA* gene

Among *Staphylococcus* spp., 18 isolates were resistant to ceftiofur. The *mecA* gene was detected in 61.1% (11/18; 95% CI: 35.75–82.70) in the ceftiofur-resistant isolates. Specifically, *mecA* gene was detected in 83.3% (5 out of 6; 95% CI: 35.88–99.58) in the ceftiofur resistant *S. aureus*, and those isolated were defined as methicillin-resistant *S. aureus* (MRSA).

In the present study, *mecA* gene was not detected in ceftiofur-negative *Staphylococcus* spp.

Discussion

The prevalence of SCM varies globally, influenced by factors such as management practices and geographical

location. The present study identified a lower prevalence indicating improved management practices in the study area than was observed in previous studies in Bangladesh (Rana et al., 2022). Risk factors such as udder cleanliness and post-milking teat disinfection were significant contributors to the occurrence of SCM. AMR patterns revealed concerning levels of multidrug resistance in *Staphylococcal* and *Streptococcal* isolates, emphasizing the need for prudent antibiotic use and further genomic surveillance investigations.

Among three different levels of prevalence including herd, animal, and quarter level, we mainly focused on animal level that was recorded 35.95% while Ndahtuye et al., (2019) reported a much higher prevalence of 76.2%. Besides, Mbindyo et al., (2020) also reported 73.1% dairy cows were affected by SCM in Kenya. However, Tezera and Aman Ali, (2021) reported a much lower rate (28.34%) of SCM in Ethiopia, while the overall cow level prevalence was 40.3%. The variation of different level of prevalence influenced by a variety of factors ranging from characteristics of individual animal to hygienic management of the herds.

The spectrum of SCM etiologic agents in the present study was aligned with a recent study by Fazal et al., (2023) in Bangladesh while differ from previous studies by Kakooza et al., (2023) in Uganda, Abebe et al., (2016) in Ethiopia, and Ferdous et al., (2019) in Bangladesh. All these previous studies reported *S. aureus* as the most common causative agent of SCM, whereas in the present study, *S. chromogenes* was the most prevalent (67.27%) agent among other bacterial species, which were similar to the results reported by Fazal et al., (2023).

The environmental risk factors associated with herd level differ from place to place. The present study has highlighted five key factors, which resulted in a significant influence on the occurrence of SCM. In contrast, a recent study by Fazal et al., (2023) in Bangladesh highlighted different risk factors, such as age, number of lactations, udder condition, and teat positioning (right-left or front-back), which were different from the current findings in the present study. In multivariate logistic regression analysis, the absence of post-milking teat disinfection showed a noteworthy association, with cows

TABLE 3. ANTIMICROBIAL SUSCEPTIBILITY PATTERNS OF *STAPHYLOCOCCUS* SPP. AND *STREPTOCOCCUS* SPP. ISOLATED FROM BOVINE SUBCLINICAL MASTITIS MILK SAMPLES IN THE PRESENT STUDY

Bacterial species	Antimicrobial Susceptibility	AMP	FOX	CIP	E	CN	P	SXT	TE
<i>S. aureus</i> (n = 9)	Sensitive (%)	0	33.3	77.8	44.5	44.5	0	66.7	44.5
	Intermediate (%)	0	0	22.2	22.2	33.3	0	11.1	44.4
	Resistant (%)	100	66.7	0	33.3	22.2	100	22.2	11.1
<i>S. chromogenes</i> (n = 37)	Sensitive (%)	48.6	67.6	54.1	37.8	73	6	59.5	62.2
	Intermediate (%)	0	0	24.3	43.3	18.9	0	16.2	18.9
	Resistant (%)	51.4	32.4	21.6	18.9	8.1	31	24.3	18.9
<i>S. simulans</i> (n = 5)	Sensitive (%)	0	100	100	60	80	0	80	100
	Intermediate (%)	0	0	0	0	20	0	0	0
	Resistant (%)	100	0	0	40	0	100	20	0
<i>S. agalactiae</i> (n = 1)	Sensitive (%)	0	100	0	0	0	0	0	100
	Intermediate (%)	0	0	100	0	0	0	0	0
	Resistant (%)	100	0	0	100	100	100	100	0
<i>S. dysgalactiae</i> (n = 3)	Sensitive (%)	0	66.7	66.7	0	0	0	0	66.7
	Intermediate (%)	0	0	33.3	66.7	66.7	0	33.3	33.3
	Resistant (%)	100	33.3	0	33.3	33.3	100	66.7	0

AMP, Ampicillin; FOX, Cefoxitin; CIP, Ciprofloxacin; E, Erythromycin; CN, Gentamicin; P, Penicillin; SXT, Trimethoprim-Sulfamethoxazole; TE, Tetracycline.

having a 3.08 times higher likelihood of SCM compared to their non-risk counterparts. This situation creates a window of opportunity for organisms to easily enter the teat canal shortly after milking, which underscores the importance of teat dipping in iodine-based solutions. Udder cleanliness emerged as a substantial factor, with dirty udders carrying a 7.83 times higher risk of SCM. This aligns closely with the findings of Abebe et al., (2016) who reported an even higher odds ratio (OR = 8.7) for acquiring SCM among cows with unclean udders. Notably, the absence or delay in feeding post-milking was associated with a substantial 9.81 times higher risk of mastitis occurrence. This phenomenon can be explained by the cow typical post-milking behavior of resting and sitting down, which leaves their teats open. This open state provides an opportunity for opportunistic microorganisms to enter the udder through the teat canal. Conversely, offering feed immediately encourages cows to remain standing while consuming feed.

The inappropriate use of antimicrobial agents to treat subclinical and clinical mastitis may lead to the development of multidrug resistance. In the present study, all *S. aureus* isolates showed resistance to ampicillin and penicillin. This trend is consistent with the findings by Fazal et al., (2023). However, ciprofloxacin exhibited noteworthy effectiveness against *S. aureus*, demonstrating susceptibility rates of 77.8% of isolates, aligning with the finding by Abed et al., (2021). Among the 37 *S. chromogenes* isolates, a substantial majority displayed susceptibility to cefoxitin (67.6%). All *S. simulans* isolates exhibited 100% susceptibility to cefoxitin, ciprofloxacin, and tetracycline, consistent with the previous findings by Kim et al., (2019).

A considerable proportion of the total isolates (21.8%) in the current study exhibited patterns of MDR. Similar findings were previously reported by Rana et al., (2022) and Dabele et al., (2021). The potential development of MDR can be attributed to the excessive use of similar groups of antimicrobial agents and the repetitive administration of antimicrobials sharing similar modes of action, as noted by Das et al., (2022).

The *mecA* gene is a critical genetic determinant associated with antibiotic resistance, specifically with resistance to beta-lactam antibiotics (Rafif Khairullah et al., 2022). In the present study, *mecA* gene was identified in 61.1% (11/18) cefoxitin-resistant isolates, while Fazal et al., (2023) detected the gene in 32% of *S. aureus* species. Moreover, Rafif Khairullah et al., (2022) reported that 53.3% of *S. aureus* carried *mecA* gene. In the present study, a relatively high percentage of cefoxitin-resistant *Staphylococcus* spp. isolates were found to be negative for the *mecA* and *mecC* genes, which differs from findings in other studies. This variation may be attributed to alternative resistance mechanisms. Alterations in penicillin-binding proteins or the overexpression of efflux pumps may contribute to cefoxitin resistance in the absence of *mecA* or *mecC*. Additionally, regional variations in AMR patterns, differences in the bacterial strains isolated, or the methods used for detecting resistance genes may also explain these findings. Previous studies have reported variations in the prevalence of these resistance genes, and the results of the current study highlight the need for further investigation of other genetic or phenotypic mechanisms contributing to

cefazolin resistance in *Staphylococcus* spp. in this region (Fazal et al., 2023; Rafif Khairullah et al., 2022).

One limitation of the current study was the use of composite milk samples from all four quarters of each cow, rather than individual quarter samples, which is recommended by the NMC guidelines. While this approach provided a broader representation of the udder health status at the cow level and the employed bacteriological methods still allowed for the accurate identification of the bacterial pathogens, it may have impacted the prevalence of bacterial pathogens, potentially leading to underestimation or overestimation of the prevalence of bacterial pathogens. The decision to use composite samples was made to streamline sample collection and analysis in field conditions, but future studies should consider quarter-level sampling for more precise bacteriological assessment. Due to limited resources, molecular characterization and genomic investigations were not performed. Further investigations involving large number of farms may be included. Moreover, our recommendation is to study the biotype pathogenic bacteria by accurate real-time technology such as matrix-assisted desorption/ionization time of flight mass spectrophotometry.

Conclusions

The present study highlights the significant prevalence of SCM in cows, emphasizing its economic and public health implications. *S. chromogenes* was identified as a prominent causative agent, with varying antibiotic resistance patterns observed. The high prevalence of MDR in the present study is concerning for both animal and public health. Key risk factors include poor udder hygiene, tick infestation, and lack of post-milking teat disinfection and dry cow therapy. Targeted interventions and improved antibiotic stewardship are essential measures to enhance disease management, and to ensure dairy sustainability, and mitigate AMR.

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Authors' Contributions

A.A.F.: Conceptualization, data curation, methodology, investigation, resources, writing—original draft, reviewing and editing. H.H.: Conceptualization, investigation, methodology, data curation, writing—original draft, reviewing and editing, data analysis, data curation. M.H.R.: Methodology, data and sample collection. M.R. and K.A.B.: Methodology, data and sample collection, writing—original draft, reviewing and editing. M.S.S., M.A., A.S., R.B., H.M.S., A.N., and

Y.A.H.: Investigation, validation, writing—original draft, reviewing and editing. M.E.Z., M.M.R. and H.B.: Conceptualization, designing, investigation, supervision, resources, software, data analysis, validation, writing—original draft, reviewing and editing. All authors read and approved the final article.

Disclosure Statement

The authors declare that they have no competing interests.

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Supplementary Material

Supplementary Table S2
Supplementary Table S3
Supplementary Table S4
Supplementary Data S1

References

- Abebe R, Hatiya H, Abera M, et al. Bovine mastitis: Prevalence, risk factors and isolation of *Staphylococcus aureus* in dairy herds at hawassa milk shed, South Ethiopia. *BMC Vet Res* 2016;12(1):270; doi: 10.1186/S12917-016-0905-3
- Abed AH, Menshawy AMS, Zeinhom MMA, et al. Subclinical mastitis in selected bovine dairy herds in North Upper Egypt: Assessment of prevalence, causative bacterial pathogens, antimicrobial resistance and virulence-associated genes. *Microorganisms* 2021;9(6); doi: 10.3390/MICROORGANISMS9061175
- Asha MN, Chowdhury MSR, Hossain H, et al. Antibacterial potential of lactic acid bacteria isolated from raw cow milk in Sylhet district, Bangladesh: A molecular approach. *Vet Med Sci* 2024;10(3):e1463; doi: 10.1002/VMS3.1463
- Busato A, Trachsel P, Schällibaum M, et al. Udder health and risk factors for subclinical mastitis in organic dairy farms in Switzerland. *Prev Vet Med* 2000;44(3–4):205–220; doi: 10.1016/S0167-5877(00)00104-5
- Chowdhury M, Rahman M, Hossain H, et al. Subclinical mastitis of buffaloes in Asia: Prevalence, pathogenesis, risk factors, antimicrobial resistance, and current treatment strategies. *J Anim Sci Technol* 2024; doi: 10.5187/JAST.2024.E66
- CLSI. Clinical and Laboratory Standards Institute (CLSI). Performance Standards for Antimicrobial Susceptibility Testing. 30th ed. CLSI Supplement M100. CLSI; 2020.
- Dabele DT, Borena BM, Admasu P, et al. Prevalence and risk factors of mastitis and isolation, identification and antibiogram of *Staphylococcus* species from mastitis positive zebu cows in Toke Kutaye, Cheliya, and Dendi districts, west shewa zone, Oromia, Ethiopia. *Infect Drug Resist* 2021;14: 987–998; doi: 10.2147/IDR.S295257
- Das T, Rana EA, Dutta A, et al. Antimicrobial resistance profiling and burden of resistance genes in zoonotic *Salmonella* isolated from broiler chicken. *Vet Med Sci* 2022;8(1): 237–244; doi: 10.1002/VMS3.648
- El-Sayed A, Awad W, Abdou NE, et al. Molecular biological tools applied for identification of mastitis causing pathogens. *Int J Vet Sci Med* 2017;5(2):89–97; doi: 10.1016/j.ijvsm.2017.08.002
- Al Emon A, Hossain H, Chowdhury MSR, et al. Prevalence, antimicrobial susceptibility profiles and resistant gene identification of bovine subclinical mastitis pathogens in Bangladesh. *Heliyon* 2024;10(14):e34567; doi: 10.1016/j.heliyon.2024.e34567
- Fazal MA, Rana EA, Akter S, et al. Molecular identification, antimicrobial resistance and virulence gene profiling of *Staphylococcus* spp. Associated with bovine sub-clinical mastitis in Bangladesh. *Vet Anim Sci* 2023;21:100297; doi: 10.1016/j.vas.2023.100297
- Ferdous H, Anjuman A, Mahfujur R, et al. Characterization of bacterial isolates, antibiogram profile and pro-inflammatory cytokines in subclinical mastitis in cross-bred dairy cows. *AJVS* 2019;62(2):1; doi: 10.5455/AJVS.58885
- Guimarães JLB, Brito MAVP, Lange CC, et al. Estimate of the economic impact of mastitis: A case study in a holstein dairy herd under tropical conditions. *Prev Vet Med* 2017;142: 46–50; doi: 10.1016/j.prevetmed.2017.04.011
- Hoque MF, Mehedi AF, Chowdhury EMS, et al. Prevalence and antibiotic susceptibility profile of *Staphylococcus aureus* in clinical and subclinical mastitis milk samples. *BJVM* 2023; 21(1); doi: 10.33109/BJVMJJ2023FAM2
- Hoque MN, Das ZC, Rahman ANMA, et al. Molecular characterization of *Staphylococcus aureus* strains in bovine mastitis milk in Bangladesh. *Int J Vet Sci Med* 2018;6(1):53–60; doi: 10.1016/j.ijvsm.2018.03.008
- Juhász-Kaszanyitzky É, Jánosi S, Somogyi P, et al. MRSA Transmission between cows and humans. *Emerg Infect Dis* 2007;13(4):630–632; doi: 10.3201/EID1304.060833
- Kakooza S, Mutebi F, Ssajjakambwe P, et al. Mastitis on selected farms in Wakiso district, Uganda: Burden, pathogens and predictors of infectivity of antimicrobial resistant bacteria in dairy herds. *Vet Med Sci* 2023;9(5):2376–2385; doi: 10.1002/VMS3.1234
- Kim SJ, Moon DC, Park SC, et al. Antimicrobial resistance and genetic characterization of coagulase-negative *Staphylococci* from bovine mastitis milk samples in Korea. *J Dairy Sci* 2019;102(12):11439–11448; doi: 10.3168/jds.2019-17028
- Krishnamoorthy P, Goudar AL, Suresh KP, et al. Global and countrywide prevalence of subclinical and clinical mastitis in dairy cattle and buffaloes by systematic review and meta-analysis. *Res Vet Sci* 2021;136:561–586; doi: 10.1016/j.rvsc.2021.04.021
- Leibovici L, Scudeller L, Kalil A, et al. Guidance on reporting multivariable regression models in CMI. *Clin Microbiol Infect* 2020;26(1):1–2; doi: 10.1016/j.cmi.2019.10.037
- Magiorakos AP, Srinivasan A, Carey RB, et al. Multidrug-Resistant, Extensively Drug-Resistant and Pandrug-Resistant Bacteria: An International Expert Proposal for Interim Standard Definitions for Acquired Resistance. *Clin Microbiol Infect* 2012;18(3):268–281; doi: 10.1111/J.1469-0691.2011.03570.X

- Mahen MSK, Chowdhury MSR, Hossain H, et al. Investigating the infection dynamics and molecular detection of *Cryptosporidium* in buffaloes in Sylhet, Bangladesh. *Vet Parasitol Reg Stud Reports* 2024;52:101043; doi: 10.1016/j.vprsr.2024.101043
- Mbindyo CM, Gitao GC, Mulei CM. Prevalence, etiology, and risk factors of mastitis in dairy cattle in Embu and Kajiado counties, Kenya. *Vet Med Int* 2020;2020(1):8831172; doi: 10.1155/2020/8831172
- Al Naser J, Hossain H, Chowdhury MSR, et al. Exploring of spectrum beta lactamase producing multidrug-resistant *Salmonella enterica* serovars in goat meat markets of Bangladesh. *Vet Anim Sci* 2024;25:100367; doi: 10.1016/J.VAS.2024.100367
- National Mastitis Council (NMC) Protocols, Guidelines and Procedures. Microbiological Procedures for the Diagnosis of Bovine Udder Infection and Determination of Milk Quality. [NMC publication, 2004]. Available at: <https://www.nmconline.org/nmc-protocols-guidelines-and-procedures/>
- Ndahetuye JB, Persson Y, Nyman AK, et al. Aetiology and prevalence of subclinical mastitis in dairy herds in peri-urban areas of Kigali in Rwanda. *Trop Anim Health Prod* 2019;51(7):2037–2044; doi: 10.1007/S11250-019-01905-2/TABLES/3
- Oliveira CSF, Hogeveen H, Botelho AM, et al. Cow-specific risk factors for clinical mastitis in Brazilian dairy cattle. *Prev Vet Med* 2015;121(3–4):297–305; doi: 10.1016/j.prevetmed.2015.08.001
- Paramasivam R, Gopal DR, Dhandapani R, et al. Is AMR in dairy products a threat to human health? An updated review on the origin, prevention, treatment, and economic impacts of subclinical mastitis. *Infect Drug Resist* 2023;16:155–178; doi: 10.2147/IDR.S384776
- Rafif Khairullah A, Rehman S, Agus Sudjarwo S, et al. Detection of *mecA* Gene and Methicillin-Resistant *Staphylococcus aureus* (MRSA) isolated from milk and risk factors from farms in Probolinggo, Indonesia. *F1000Res* 2022;11:722; doi: 10.12688/f1000research.122225.3
- Rahman MdM, Hossain H, Chowdhury MdSR, et al. Molecular characterization of multidrug-resistant and extended-spectrum β -lactamases-producing *Salmonella enterica* serovars enteritidis and typhimurium isolated from raw meat in retail markets. *Antibiotics (Basel)* 2024;13(7):586; doi: 10.3390/ANTIBIOTICS13070586
- Rana EA, Fazal MA, Alim MA. Frequently used therapeutic antimicrobials and their resistance patterns on *Staphylococcus aureus* and *Escherichia coli* in mastitis affected lactating cows. *Int J Vet Sci Med* 2022;10(1):1–10; doi: 10.1080/23144599.2022.2038494
- Sarker SC, Parvin MS, Rahman AKMA, et al. Prevalence and risk factors of subclinical mastitis in lactating dairy cows in north and south regions of Bangladesh. *Trop Anim Health Prod* 2013;45(5):1171–1176; doi: 10.1007/S11250-012-0342-7
- Schadt I. Health concerns about possible long-term effects of legally marketed milk and dairy from animals with intramammary infections. *Front Public Health* 2023;11:1200924; doi: 10.3389/FPUBH.2023.1200924/FULL
- Shome BR, Das Mitra S, Bhuvana M, et al. Multiplex PCR assay for species identification of bovine mastitis pathogens. *J Appl Microbiol* 2011;111(6):1349–1356; doi: 10.1111/J.1365-2672.2011.05169.X
- Tezera M, Aman Ali E. Prevalence and associated risk factors of bovine mastitis in dairy cows in and around Assosa Town, Benishangul-Gumuz regional state, Western Ethiopia. *Vet Med Sci* 2021;7(4):1280–1286; doi: 10.1002/VMS3.454

Address correspondence to:
Md. Mahfujur Rahman, DVM, MS, PhD
Department of Medicine
Faculty of Veterinary
Animal and Biomedical Sciences
Sylhet Agricultural University
Sylhet 3100
Bangladesh

E-mail: mahfuj.vetmed@sau.ac.bd

Mohamed E. El Zowalaty, MS, PhD
Department of Microbiology and Immunology
Faculty of Pharmacy
Ahran Canadian University
Giza 3221405
Egypt

E-mail: elzow005@gmail.com